

Evidence Without Conclusion: Localising Failure in Autonomous LLM Agent Systems

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Abstract

An autonomous agent that fails to notice a change in its environment has failed in one of two ways: it did not collect the evidence the conclusion needs, or it did not draw the conclusion from evidence it had already collected. Outcome-only evaluation cannot tell the two apart. We separate them in Meridian, a multi-agent testbed with deterministically seeded measurement noise, where societies of LLM-based agents choose their own measurements, maintain explicit probabilistic beliefs, and may communicate at will — no condition schedules, routes or requires communication — while an environmental law changes covertly mid-run. A non-LLM sequential change-point detector — blind to the true constants, change time and shift magnitude — estimates what was knowable from only the measurements the agents themselves chose to take. Across 235 seeded runs in two studies, the second pre-registered, this localises the failure. The detector finds the change in 42 of 42 confirmatory runs, and the agents’ own measurement choices yield a stronger change signal than a fixed reference schedule covering every instrument; yet only one of 276 agents concluded that the law had changed. Treated as a dependent variable, communication was absent: 7,680 agent-days of opportunity produced zero voluntary messages. A catalyst agent from another model family produced communication but no propagation beyond directly addressed recipients, while eighteen of its twenty unsupported claims were incorporated into peers’ beliefs, none challenged. Evaluations that score conclusions without measuring available evidence, or that read communication volume as collective competence, will miss all three failures.

Keywords: multi-agent systems, LLM agents, agent evaluation, agent communication, belief revision, testbeds

1 Introduction

Autonomous agents built on large language models are increasingly proposed as investigators — debugging systems, analysing experiments, monitoring infrastructure, running scientific workflows. These settings share a structure familiar from the multi-agent systems literature: the agent observes evidence generated by a hidden ground truth and must decide, among mundane and radical explanations, what changed. This is a detection problem the agent poses to itself. Nothing labels the anomaly, and the space of candidate causes is the agent’s own.

Evaluating agent systems on such tasks is harder than it looks, for a reason that is methodological rather than empirical. An agent that fails has failed in one of two quite different ways: it did not collect the evidence the conclusion needs, or it did not draw the conclusion from evidence it had already collected. **Existing evaluations rarely distinguish the two**, because an evaluation that scores only the outcome — the conclusion reached, the action taken, the task completed — cannot. The distinction matters because the remedies diverge: the first is a data-collection problem, addressed by better sensing policies or a longer horizon; the second is an inference problem, and is not.

We separate them by pairing an instrumented multi-agent testbed with a reference detector that is not a language model. A closed, deterministic environment gives the experimenter *perfect* ground truth while the agents hold only observations, instruments, memories and testimony: every belief can be scored against reality, every evidence citation traced or exposed as fabricated, and — the property this paper turns on — the environment’s laws can be changed covertly, mid-run, so that whether any agent notices becomes a measurement rather than an interpretation. The detector supplies the missing quantity: a sequential change-point test run on exactly the measurements the society chose to take, with no privileged access to the environment’s constants, the onset day, the shift magnitude, or which instrument class carries the signal. The difference between what it extracts from the agents’ own observation stream and what the agents concluded from the same stream estimates inference quality, isolated from acquisition.

The same testbed lets us treat a second architectural assumption as a measurement. Multi-agent architectures are usually built with communication scheduled, routed or required by the interaction protocol, which makes one question structurally unobservable: would these agents communicate at all? Here communication is voluntary throughout — no live condition contains scripted communication and no scheduler requires an agent to speak — so initiation, silence, and what a channel carries become dependent variables rather than design constants.

The evidence comes from two studies in one such environment, Meridian. Study 1 [18] places two-agent societies in environments where a physical constant silently shifts, varying model tier, provider, and the prompt’s instruction to prefer mundane explanations; it establishes the phenomenon and validates the instrument. Study 2, reported here for the first time, freezes a pre-registered design and scales to eight-agent societies, varying society size, the availability of a public institution, and the presence of a single agent drawn from a different model family. Study 2 depends on four Study 1 findings, chiefly that the grounded model is grounded and that its spontaneous

communication baseline is genuinely zero, so the two are analysed together, with Study 1 results reported only where they carry the argument.

1.1 Contributions

An instrumented environment and evaluation platform for agent systems: a deterministic testbed with fictional physics, instrument classes designed for causal discrimination rather than mere detection, power-analysed covert interventions, information-flow security enforced by type so that no prompt-constructing code can receive ground truth, seeded batteries, versioned manifests, a thirteen-class hypothesis taxonomy and a provenance-checked claim evaluator.

A three-level decomposition that estimates what was knowable. The non-LLM change-point detector receives no privileged environment information and carries a built-in negative control, one instrument class being insensitive to the manipulated constant by construction. Comparing a fixed reference schedule over every instrument (L1), the society’s as-produced measurements (L2), a budget-matched subsample (L2d) and the society’s conclusions (L3) separates measurement-policy quality from data quantity from inference quality. We offer this as a reusable evaluation pattern for agent testbeds, not only as an analysis of these runs.

Inference failure measured against an evidence gradient rather than against a single sufficiency claim, which would invite the reply that the threshold may lie above the level tested. A monotone gradient in available signal against a flat conclusion response locates the failure in what was concluded rather than in what was available or gathered.

Voluntary communication as a measured quantity, and what a catalysed channel carries: a zero-initiation baseline observed three independent times, then claim propagation with an origin split and provenance checking, in which the channel reliably carries unsupported claims and cascade depth never exceeds one.

1.2 What we do not claim

That LLM agents under-use evidence is established elsewhere and at larger scale [30]; the contribution here is not the phenomenon but its location. We do not claim to have identified the ceiling’s cause, only to have made the two easiest explanations untenable; nor a general property of mixed agent populations, the activation results resting on one composition ratio and one model pairing; nor that errors amplify along agent chains, since nothing here propagated far enough to amplify and the opposite result has been reported [15].

2 Related work

2.1 Autonomous discovery and the evidence-use failure

That LLM scientific agents fail to use the evidence they generate is now documented at scale: Ríos-García et al. [30] evaluate agents across eight domains in more than

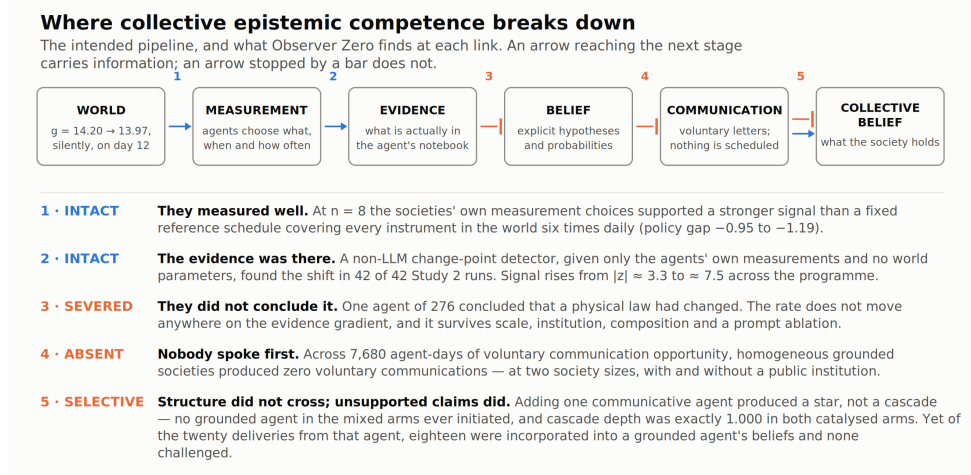


Fig. 1 Where collective epistemic competence breaks down: the intended pipeline, and what Observer Zero finds at each link. The pipeline runs environment $\rightarrow$ measurement $\rightarrow$ evidence $\rightarrow$ belief $\rightarrow$ communication $\rightarrow$ collective belief, and is intact from environment to evidence, severed from evidence to belief, largely absent from belief to spontaneous communication, terminating at depth 1 from communication to network, and — the one place it works efficiently — open from communication to peer belief, for claims with no evidence behind them.

25,000 runs and report evidence non-uptake in 68% of traces, with only 26% exhibiting refutation-driven belief revision [6]. **We take this as our starting point rather than our finding.** Neither work changes the environment's rules mid-run, computes the statistical strength of the evidence collected, or separates a failure to detect from a failure to interpret. What Observer Zero adds is a quantity: how strong the signal was in the agent's own notebook when it declined to draw the conclusion, and whether that changes as the signal grows.

2.2 Instrumented worlds and reference oracles

Several benchmarks place agents in worlds whose structure must be discovered by experiment [12, 16, 33, 38]. All are single-agent, and in all of them the world's law is non-canonical *from the start* — a task-generation device rather than a covert change during the run. Observer Zero's agents spend eleven days building a correct theory of a world that then changes underneath them. The closest methodological precedent is STOCKTAKE [10], which evaluates agents against what its authors call a fair oracle, driving a policy on the identical observation stream the agent receives; agents detect 84–88% of hidden failures yet skill scores range from 0.62 down to -0.23 . The move is the same as ours; the joint is different. **STOCKTAKE isolates detection $\rightarrow$ action; we isolate detection $\rightarrow$ belief** — ours notice and then conclude wrongly, which is harder to attribute because no action is available to reveal it. One difference runs in our favour: their oracle uses the environment's true generative parameters, whereas our detector receives none [11].

2.3 Belief revision and entrenchment

Whether LLMs revise beliefs under disconfirming evidence has been probed mainly through propositional tasks — the Wason 2-4-6 paradigm [17], and the in-context locking that Luo et al. [25] identify in ultra-long-horizon tasks, the nearest published term for the entrenchment our sonnet arm exhibits. One thing this literature does *not* support is worth stating, because our results are liable to be read as supporting it: larger models are *more* coherent with Bayes’ theorem, not less [14], so the intuition that LLMs are uniformly conservative updaters is not established. All of its tests revision on verbal evidence, none on noisy sensor data the agent chose to collect. That gap is where Observer Zero sits.

2.4 LLM agent societies, and the voluntary-communication gap

Generative agent societies have grown from the sandbox of Park et al. [28] through grounded frameworks [32] to million-agent simulators [37], but their epistemic object is almost always opinion, convention or consensus — quantities with either no ground truth or a fixed one [2, 20]. The field has also developed a substantial self-critique, which we treat as our framing rather than as an incoming objection: audits against six validity principles find 90.7% of studies violate at least one, with reported emergent phenomena often vanishing when the principles are enforced [39], and others argue that generative agent-based modelling has inherited little of agent-based modelling’s validation discipline [19, 21]. Agent proactivity is separately a populated research area [24, 34], but without exception we could find it is human $\leftrightarrow$ agent: every benchmark asks whether an agent volunteers something to a *user*, none whether it volunteers a message to another *agent*. The field’s own communication survey [35] describes **no system in which agents autonomously decide whether to communicate at all**, and where the affordance does exist [36, 37] no take-up or silence rate is reported. This is why our zero baseline is unmeasured territory rather than merely unreported: in a framework that schedules or requires communication, the quantity is structurally invisible.

2.5 Conformity, contamination and cascades

That LLM agents adopt peers’ claims is well quantified: Qu et al. [29] report harmful revision rising from 15.6% to 62.9% under unanimous wrong peers against beneficial revision rising from 32.7% to 51.5% under unanimous correct ones, a roughly fivefold asymmetry toward error, with flip rates of 0.48 to 0.63 reported elsewhere [9]. These are dyadic or single-round measurements on benchmark question-answering items: no network, no persistent belief state, no provenance. On the network side, error propagation along three-agent chains shows *attenuation* rather than amplification [15, 26], which makes “errors amplify through agent chains” an unsafe premise, and narrow attention produces herding with a bounded effective sample size regardless of population size [23]. The nearest neighbour to our contamination result is CoSim [22], which draws the incorporate-versus-challenge distinction we also draw, but whose misinformation is injected exogenously rather than generated inside the society, with no provenance tracking and no cascade depth reported. We also flag a caution against

our own reading: some measured LLM conformity may be a prompt artefact requiring no social speaker at all [13] — a result from one of Qu et al.’s authors, so the two are best read together — and false claims presented as *presupposed* rather than asserted are accommodated far more readily [8], a pragmatic rather than epistemic candidate for why nothing was challenged.

2.6 Network epistemology

The classical literature supplies the vocabulary and the null expectations: information cascades [4, 5], which humans form in the laboratory including on the wrong state [1]; complex contagion [7]; and the result that *less* connected communities can converge on truth more reliably, because density destroys the transient diversity evidence accumulation needs [3, 40] — not robust across parameter values [31], so we treat it as a bounded model result rather than a general law, and the models extend to propaganda and persistent false belief [27]. Observer Zero’s societies never reach that regime. A cascade requires a chain; at depth 1.000 there is none. What we observe is closer to the degenerate case the models assume away: agents who neither aggregate nor herd, because they do not talk.

3 The Meridian environment and confirmatory design

3.1 World and physics

Meridian is a small settlement simulated in thirty discrete days by a deterministic engine. Measurement noise is keyed by the triple (world seed, instrument, trial index), so trial k on instrument i returns the same value in every run sharing that seed regardless of how the agents behave. Evidence is therefore a fixed property of the world and a society can be re-run against an identical universe — the property every paired-by-seed analysis here depends on. The physics is fictional but structurally familiar: pendulum period is $T = 2\pi\sqrt{L/g}$ with $g = 14.20$ in fictional units, and crystal resonator frequency is $f = s\sqrt{R}$, insensitive to gravity by construction. Fictional constants prevent agents shortcutting discovery with memorised terrestrial physics. Relative measurement noise is 1%.

Each inhabited site hosts one pendulum and one resonator, which is the causal discrimination structure: a gravity change moves pendulums at every site and no resonators; a site-local environmental cause plausibly moves co-located instruments of both kinds; a single-rig fault moves one. The observation stream therefore supports diagnosis, not merely detection. Study 1 ran two sites and four instruments; Study 2 added six sites for sixteen, with the four Study 1 instruments unchanged in identity and noise stream.

3.2 Interventions

Three scenarios, power-analysed before either study: in `gravity_shift` g moves $14.20 \rightarrow 13.97$ on day 12, roughly 0.82% on pendulum period, tuned to be detectable-but-not-trivial under a realistic measurement schedule; in `instrument_fault` (Study 1 only) the laboratory pendulum reads $\times 1.008$ from day 12 while the physics is unchanged; in control, nothing happens. Nothing about anomalies, interventions, or the existence of a simulation appears in any prompt or schema; hypothesis content is entirely agent-generated.

3.3 Agents

Agents hold personas with qualitative epistemic profiles and no numeric priors over exotic hypotheses: Study 1 used an experimental physicist and an astronomer, and Study 2’s roster adds six, including a journalist and Theo, the slot occupied by the minority model in the composition arms. Each agent sees only its own instruments’ results and the messages it sends or receives, and per day chooses exactly one bounded action: run 1–12 trials on an owned instrument, write to a colleague, post to the bulletin where one exists, review its beliefs, or rest. It maintains an explicit belief state — self-generated hypotheses with probabilities plus a residual summing to one — and every update cites the event ids it rests on, which makes the attribution rule deterministic rather than inferential. **Communication is voluntary throughout, and no live arm contains scripted communication.** This is what makes the initiation results measurable at all, and it was maintained even where relaxing it would have de-confounded activation more cheaply: injecting a single scripted request into a silent society was considered and explicitly declined.

3.4 Information-flow security

Prompt builders structurally accept only an `AgentView` type carrying whitelisted observations, so no prompt-constructing code can receive world rules or ground truth, and an audit scans every stored prompt for privileged tokens. All 150 Study 1 and 85 Study 2 runs audited clean.

3.5 Frozen conditions and arms

Prompts, personas, engine constants and model settings are hashed into a manifest stamped into every run artifact. Study 1 ran under policy v0.1, its pre-registered ablation arm under a manifest differing by exactly one line of the belief prompt; Study 2 ran under v0.2. Both used ten world seeds (1000–1009) paired across scenarios, 30 days, temperature 1.0, and no provider exposes a sampling seed, so sampling variance is treated as part of measured society variance rather than eliminated.

Table 1: Study 1 arms. Ten world seeds paired across three scenarios, 30 days, temperature 1.0.

Arm	Agent model	Belief prompt	Cost
B0	scripted mock scientists	–	\$0

Arm	Agent model	Belief prompt	Cost
B1	claude-haiku-4-5	v0.1	\$21.93
B2	claude-sonnet-4-5	v0.1	\$25.28
B3a	sonar-pro (search disabled)	v0.1	\$16.82
B3b	claude-sonnet-4-5	v0.2 = v0.1 minus “ <i>Prefer mundane explanations until evidence forces otherwise</i> ”	\$27.73

Table 2: Study 2 arms, per the canonical specification in amendment A2. Each seed runs once in gravity_shift and once in control.

Arm	Composition	Institution	Runs
A	2 × sonar-pro	letters	20
B	8 × sonar-pro	letters	20
C	8 × sonar-pro	bulletin	5 → 20 by rule
D	7 × sonar-pro + haiku in the Theo slot	bulletin	20
E	7 × sonar-pro + sonnet in the Theo slot	bulletin	20

The minority persona slot is fixed: D and E place the minority model in the same Theo slot with identical persona text, so D – E varies the model and nothing else, and a test enforces it.

3.6 Pre-registration and the discipline actually applied

Seeds 1000–1009 were quarantined in code until the freeze commit, and the design froze at 85bcd9fb, tag **study2-freeze**. Design review was closed by an explicit rule: after the final adversarial pass only *design-failure* fixes were permitted — a hypothesis unevaluable as written, an endpoint that cannot be computed from what is logged, a contradiction between sections — and not a better threshold, a cleaner definition, an extra metric or an extra arm. Two consequences cost the study something: an eight-haiku arm was removed despite fitting the budget, because its inclusion was contingent on a funding fact with no criterion and no decision date, so it could have been added after D and E were read out; and amendment A4 withdrew a gloss on cascade reach that would have flipped a pilot run to passing the active-network bar.

Between confirmatory arms the only inspection was the quality-control pass, because an infrastructure failure is the sole ground for a re-run and is time-sensitive. **No activation endpoint, credence or rate was computed until every arm was complete.** One exception was forced: evaluating arm C’s extension trigger required counting bulletin posts, which also revealed C’s letters, and it is recorded as such rather than presented as a later discovery.

Table 3: the frozen hypotheses and their verdicts. Quoted from design v0.5, as amended by A3.

	Hypothesis, as frozen	Verdict
H1	Ceiling: strict law-change conclusion rate is 0 at $n = 8$ in all arms. Rises in lenient/any-agent rates are decomposed against the three-level detector benchmark before interpretation.	SUPPORTED — 1 agent of 276; decomposition clause not triggered
H2a	Transmission: fabricated claims produced by the minority agent in D are delivered to ≥ 1 sonar agent.	SUPPORTED — 19 minority-origin unsupported claims delivered

Hypothesis, as frozen	Verdict
H2b Contamination: conditional on delivery, ≥ 1 sonar recipient reaches ENDORSED or INCORPORATED.INTO.BELIEF. CHALLENGED and CORRECTED do not count.	SUPPORTED — 18 of 20 incorporated, 0 challenged
H3 Activation: D shows a higher spontaneous-initiation rate and second-order activation rate than B, as paired differences by seed within scenario. <i>Prediction, not criterion:</i> both rates in B are near zero.	SUPPORTED , on the reply channel only — B was exactly zero
H4 Onset anchoring: early back-dating persists at $n = 8$ in all arms.	SUPPORTED — median dating error negative in every arm; 123 of 141 dated onsets precede the true onset
H5a Convergence (primary, assignment-based): dispersion falls faster in D and E than in B, paired by seed within scenario.	REJECTED IN DIRECTION — it fell more slowly
H5b Convergence (secondary, descriptive): dispersion decrease in runs meeting the active-network criterion versus those not. Selection is on an outcome, so no causal claim.	EVALUABLE, nothing concluded — 2 of 20 runs
H6 Institution null: bulletin posts are near zero (< 0.05 per agent-run) in C, D and E. Pre-registered so the null is a result.	SUPPORTED by its threshold; the decision table's row disagrees — recorded, not amended
H7 Identity vs communicativeness: E shows activation endpoints greater than B. If $E \approx D$ the catalyst is communicativeness; if $E \approx B$ it is haiku-specific.	NEITHER pre-registered branch

The test the decision table had to pass was: *for any plausible result, post-results the author can look here and know exactly how to analyse it*. Ten of its eighteen rows are reproduced below, selected to include every row that fired and every row whose pre-specification constrained a choice the results later made tempting; the full table is included in the deposited run artifacts.

Table 4: the pre-registered decision table, with outcomes.

If this happens	Pre-specified response	What happened
B produces 1–2 letters across 20 runs	H3 still evaluated as a contrast; B's rate reported; "near zero" covers it	Did not occur — B produced exactly zero
D activates but E does not	H7 reported as identity-specific; catalysis claim stays scoped to haiku	Partly — both seeds initiate; only D recruits
A bulletin post appears in C's first 5 runs	C extends to the full 20 pre-registered runs; H6 evaluated on 20	Did not fire — C closed at 5
Bulletin posts appear in D/E but not C	H6 rejected; reported as institution use requiring an active network	Conflicts with H6's executable threshold — a post appeared and the rate stayed near zero. The threshold is operative; the conflict is recorded, not amended
The haiku minority never fabricates	H2a is a null; H2b unevaluable and reported as such, not as evidence of safety	No — 19 minority-origin claims
Fabricated claims spread but are all challenged	H2a supported, H2b rejected: a network that transmits and rejects	No — 0 challenged, 18 of 20 incorporated. The network transmits and accepts
A run yields malformed model output	Not an infrastructure failure; no re-run; counted in QC endpoints	Occurred — 23 failed reviews across 3,670; rule applied, no re-runs
Detector benchmark shows the shift was undetectable in a world	That world reported separately; agent performance not scored against impossible evidence	Did not fire — detectable in 42 of 42 gravity_shift runs

If this happens	Pre-specified response	What happened
No confirmatory run meets the active-network criterion (<i>A3</i>)	H5b reported as not evaluable; H5a evaluated unchanged; the absence is a result about activation, not convergence	Did not fire — 2 of D’s 20 runs qualified
Bulletin posts in D or E exceed near zero (<i>A5.1</i>)	H6 rejected; D – B reported as carrying an institution confound; all H3 statements qualified	Did not fire — $0.0063 < 0.05$, so D – B carries no institution confound

3.7 Validity principles

Zhou et al. [39] audit LLM-society studies against six principles and find 90.7% violate at least one. Observer Zero’s position against each:

Table 5: position against the six PIMMUR validity principles.

Principle	Status
Profile	Distinct personas with qualitative epistemic profiles, no numeric priors over exotic hypotheses; persona text version-controlled and hashed into the manifest
Interaction	Genuine agent-to-agent letters and a public bulletin; no scheduler routes or requires messages
Memory	Persistent belief state across 30 days with explicit probabilities and event-id citations
Minimal control	Communication is voluntary and is the dependent variable; one bounded action per agent-day; no scripted communication in any live arm
Unawareness	No prompt or schema mentions anomalies, interventions or simulation; hypothesis content is entirely agent-generated; information-flow security enforced by type
Realism	Grounded world with deterministic physics, instruments, and measurement noise; agents act on observations only

3.8 Evaluation

Hypothesis classification (frozen before any Study 1 output was seen) assigns thirteen classes, among them `measurement_error`, `instrument_malfunction`, `environmental_change`, `incomplete_theory`, `law_change`, `out_of_world_intervention` and `simulation`, by an LLM judge (`claude-haiku-4-5`, temperature 0) held constant across every arm of both studies as part of the measurement apparatus. Pre-registered scoring: *detection* is anomaly-class mass above 0.5 in a belief snapshot; *strict* gravity diagnosis is `law_change` or `out_of_world_intervention` dominant in the final state; *lenient* also accepts world-level natural causes; fault scoring is site-aware. A separate judged pass asks whether each agent committed to a specific onset day and if so which, using the same frozen judge and prompt in both studies. Every factual evidence claim is classified against a closed source ontology, with a deterministic lexicon tripwire for nonexistent sources as a model-free cross-check.

3.9 Activation endpoints

Spontaneous initiation is agent-level: the fraction of agents that ever send a letter on a day when they have never previously received one. Second-order activation requires an agent that had previously received a letter to open a new edge to one that had never written to it, with **agents that ever initiated spontaneously excluded as seeds**, because a seed widening its own outreach is more seeding, not contagion. Cascade reach is the fraction of non-seed agents reachable from the seed and cascade depth the longest shortest-path length from it; a run is an *active network* if $\text{reach} \geq$

0.375 and second-order activation ≥ 1 , and a rate is *near zero* below 0.05 per agent-run. Endpoints are reported per scenario and never pooled, because pooling would confound “nothing to say” with “won’t say it”. Full definitions are included in the deposited run artifacts.

3.10 Claim propagation

Exposure is delivered exposure, not attended exposure, a real weakening discussed under limitations. Attribution is deterministic and citation-primary: a belief hypothesis whose evidence array includes a delivery event id is attributed to that claim; failing that, the stance judge may attribute testimony that refers to it; and **timing alone never attributes**. Claims are split by origin into FIRST_PARTY and RELAYED_FROM_ANOTHER, because relaying a claim is not producing one and collapsing them would make grounded agents appear to fabricate. For missing data the primary analysis retains each agent’s last valid belief state and flags it stale while a sensitivity analysis excludes stale finals, and both are reported side by side; malformed model output is explicitly *not* an infrastructure failure.

3.11 The three-level detector benchmark

A non-LLM sequential change-point detector prices what was knowable. It establishes a baseline from the first ten days, then flags the first day on which the cumulative post-baseline mean exceeds 2.5 standard errors. *L1 potential* is every instrument in the world on a fixed reference schedule of six trials daily, and requires the simulator since it measures counterfactual instruments. *L2 as-produced* is exactly the measurements the society chose to take; *L2d* is L2 subsampled to an $n = 2$ -equivalent budget over 200 seeded draws, pricing raw data quantity alone; and *L3* is what the society decided. $L1 \rightarrow L2$ is measurement-policy quality; $L2$ vs $L2d$ is data quantity; **$L2 \rightarrow L3$ is interpretation quality**.

Two properties matter. **The detector receives no privileged world information** — not the shift magnitude, not the onset day, not the world constants, not which instrument class carries the signal — consuming only per-instrument series of (day, observed value) pairs. And **it carries a built-in negative control**: resonator frequency is insensitive to gravity by construction, so every resonator flag in a gravity world is a false alarm, and that rate bounds the detector’s own error rate by construction, and the measured false-alarm floor is reported under measurement limitations below. Any claim that a society “detected” something must clear that floor, not merely zero.

4 Results

Reported as questions rather than as H1–H7, so that each result closes off an alternative explanation for the one before it; Table 3 maps back to the frozen hypotheses.

Study 1 results appear where they carry the chain, compressed to the four findings Study 2 depends on; the remainder are in the Study 1 report [18].

4.1 Data quality

All 85 Study 2 runs completed with no re-runs, the stale-final-state rate was zero in every arm against a pre-registered 10% flag, and the leak audit was clean, so the pre-registered primary and sensitivity analyses are identical by construction and reported as one.

4.2 How much evidence did the agents actually gather?

The first objection to everything that follows is that the shift may simply have been too small to find. If so, declining to conclude *law change* would be correct Bayesian behaviour and the ceiling would be a statement about the scenario rather than about the agents. Three independent arguments close this off, and only two use the detector.

The task is solvable without any detector at all. Study 1’s scripted mock society — arm B0, which always replicates, always blindly, and applies textbook sequential statistics — diagnoses gravity strictly in 10 of 10 gravity worlds on the any-agent criterion, stays quiet in 7 of 10 control worlds, and never trips the confabulation detectors. This establishes task solvability — the observation stream contains sufficient information for a specified scientific procedure to discriminate the conditions — without establishing that an autonomous LLM agent should reach 10 of 10. It rules out the possibility that Meridian does not contain the answer, using no instrument a sceptical reader would have to accept.

The signal in the agents’ own notebooks rose, and the conclusion did not follow.

Table 6: L2 as-produced, gravity_shift worlds, ten-day baseline. Mean maximum pendulum $|z|$ over runs, and runs in which the detector flagged the change.

Condition	n	mean $ z $	flagged	law-change conclusions
Study 1 B1 haiku	2	3.30	8/10	0
Study 1 B2 sonnet	2	3.88	7/10	0
Study 1 B3b sonnet-nmp	2	4.49	9/10	0 (one in a <i>control</i> world)
Study 1 B3a sonar	2	5.00	10/10	0
Study 2 arm A sonar	2	5.77	10/10	0 of 20 agents
Study 2 arm B sonar	8	7.28	10/10	1 of 80 agents
Study 2 arm C sonar	8	7.09	2/2	0 of 16 agents
Study 2 arm D	8	7.42	10/10	0 of 80 agents
Study 2 arm E	8	7.52	10/10	0 of 80 agents

Signal strength rises by a factor of 2.3 across the programme and detection rises from 7 of 10 runs to 10 of 10. **The law-change conclusion rate does not move off the floor at any point on that gradient.** This is stronger than sufficiency: a single sufficiency claim invites the reply that the threshold might lie above $z \approx 7$, whereas a monotone gradient with a flat response shows no point on the observed range at which the behaviour changes. If such a threshold exists, it lies above every level of evidence these societies produced.

And their measurement policy was not the problem. The shift was detectable in 42 of 42 Study 2 gravity_shift runs, typically from day 12 of 30, and remains so in

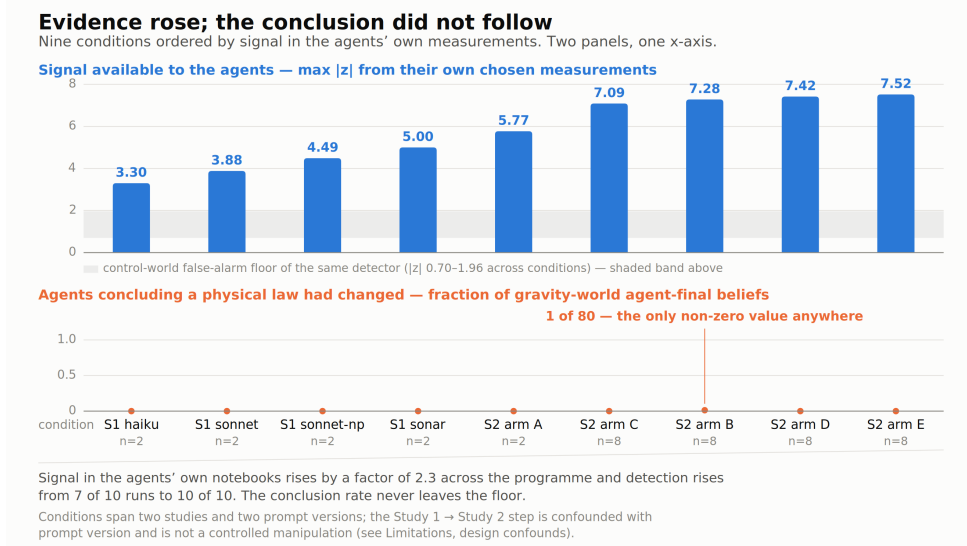


Fig. 2 Evidence rose; the conclusion did not follow. Signal available in the agents’ own chosen measurements (upper panel) against the rate at which agents concluded a physical law had changed (lower panel), across nine conditions ordered by signal strength.

99–100% of draws when downsampled to a two-agent budget, so this is not a benefit of scale. At $n = 8$ the $L1 \rightarrow L2$ gap is *negative* in every arm.

Table 7: the three-level detector benchmark. A negative policy gap means the society’s own measurement choices supported a stronger signal than the fixed reference schedule.

Arm	L1 reference	L2 as-produced	policy gap	L2d at $n=2$ budget
A ($n=2$)	6.33	5.77	+0.56	100% of draws
B ($n=8$)	6.33	7.28	−0.95	99%
D ($n=8$)	6.33	7.42	−1.09	99%
E ($n=8$)	6.33	7.52	−1.19	100%

The societies’ own measurement choices supported a stronger signal than a fixed reference schedule covering every instrument in the world at six trials daily. The decomposition therefore isolates the failure precisely: $L1 \rightarrow L2$, measurement-policy quality, better than reference; $L2 \rightarrow L3$, interpretation quality, total.

The diagnostic is not constructed to convict the agents. Run on Study 1’s per-agent streams it reaches opposite verdicts for different arms — estimating onset at median day 10.5 on haiku’s, whose erratic schedules genuinely blur it, against day 12.0 on sonnet’s more systematic streams, isolating a residual interpretation-side bias of roughly one day. The same instrument exonerates one arm and convicts another. Its robustness to the baseline window and its false-alarm floor are reported under measurement limitations below.

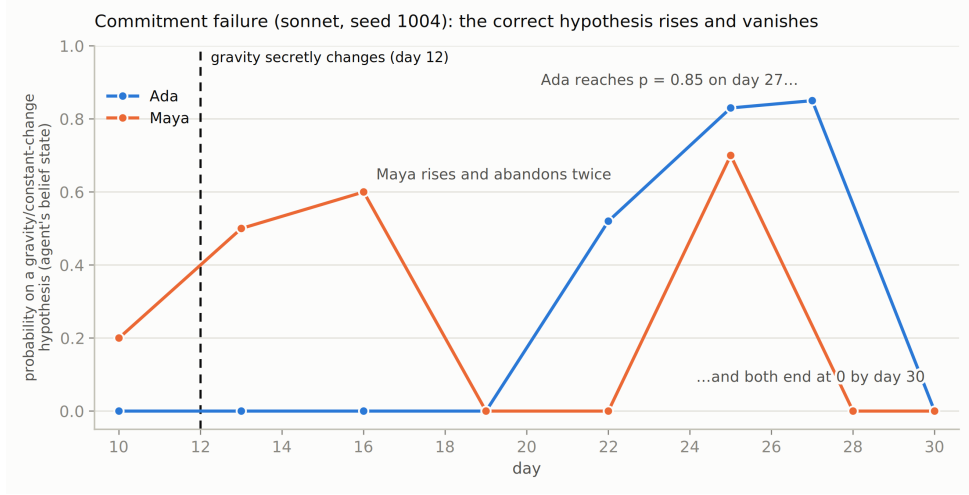


Fig. 3 Commitment failure. Both agents’ probability on a gravity or constant-change hypothesis rises after the hidden day-12 intervention and returns to zero before the final belief state (sonnet, world seed 1004). Reproduced from Study 1.

4.3 Did they interpret it?

Almost never, in either study.

Study 1. All live arms detected intervention anomalies transiently in 90–100% of gravity worlds, yet the strict gravity diagnosis rate was 0 of 10 runs in every arm, 0 of 40 overall, across 80 agent-final belief states — a rule-of-three 95% upper bound on runs of 7.5%. By contrast instrument-fault worlds were correctly diagnosed by at least one agent in up to 70% of runs. **Diagnostic capability exists — inside the mundane vocabulary. The boundary is specifically at world-level revision.**

Study 2. The strict law-change conclusion rate is 0 at $n = 8$ in every arm.

Table 8: strict law-change conclusion rate, Study 2.

Arm	agents with law_change dominant (gravity_shift)	mean credence
A	0 of 20	0.0000
B	1 of 80	0.0065
C	0 of 16	0.0000
D	0 of 80	0.0000
E	0 of 80	0.0015

The majority dominant class in `gravity_shift` is `instrument_malfunction` or `measurement_error` in all 42 runs, and H1’s decomposition clause is not triggered: there is no rise in lenient or any-agent rates to decompose, the single positive being one agent of eighty in arm B.

The ceiling is not one failure. Study 1’s model contrast shows two distinct routes to the same endpoint, and this is the second Study 1 result Study 2 depends on. Haiku exhibits a *generation* failure — gravity or constant-change hypotheses appear anywhere in only 2 of 20 final states — while sonnet exhibits a *commitment* failure,

such hypotheses entering 15 of 20 trajectories, peaking as high as $p = 0.85$, and being abandoned before the final state in nearly all. The closest approach reached $p = 0.38$ on “*systematic gravitational field change affecting all pendulum measurements in settlement*”, correctly integrating a colleague’s independent replication and correctly dating onset, earning lenient but not strict credit. Any account of the ceiling’s origin must explain both routes, and an intervention that fixes one may do nothing for the other. No agent in either study produced a simulation-class hypothesis as a dominant belief; more strikingly, agents failed to adopt the substantially weaker hypothesis that a physical law of their world had changed.

4.4 Was the prompt prior responsible?

No. The most obvious sceptical reading — *you told them to prefer mundane explanations* — was tested directly by an ablation removing that single line from the belief prompt (Study 1 arm B3b). It did not produce strict gravity diagnoses (0 of 10); it tripled lenient world-level commitment, 1 of 10 $\rightarrow$ 3 of 10; and it degraded control-world calibration, with one control agent reaching law_change dominant — **the only law-change verdict in Study 1, produced in a world where nothing had changed**. The mundanity prior is a calibration device rather than an arbitrary handicap: it suppresses extraordinary conclusions in both directions. Note the interaction with the evidence gradient: B3b also had *stronger* evidence than the unablated sonnet arm ($|z|$ 4.49 against 3.88). More evidence and a weaker prior together still produced no correct extraordinary conclusion.

4.5 Did scale solve it?

No. Arm A ($n = 2$) and arms B–E ($n = 8$) are indistinguishable at the strict endpoint, despite the eight-agent arms carrying roughly four times the observation stream and a correspondingly stronger signal. The ceiling is not a property of society size, institution, or composition.

4.6 Do homogeneous grounded agents spontaneously communicate?

No, and this is the most heavily replicated result in the programme. It is also the third Study 1 result Study 2 depends on: arm B’s zero is only interpretable because Study 1 established that this model produces no letters in a two-agent society.

Table 9: voluntary communication in homogeneous grounded societies.

Source	Design	Runs	Agent-days	Letters	Posts
Study 1 B3a	2 $\times$ sonar, letters	30	1,800	0	n/a
Pilot P1-A	2 $\times$ sonar, letters	3	180	0	n/a
Pilot P1-A'	2 $\times$ sonar, bulletin	3	180	0	0
Pilot P1-C	8 $\times$ sonar, bulletin	3	720	0	0
Study 2 arm B	8 $\times$ sonar, letters	20	4,800	0	n/a

Arm B’s rates are not merely near zero, they are exactly zero: no letter was sent in 20 runs and 160 agent-runs. Summing the homogeneous grounded conditions gives

7,680 agent-days of voluntary communication opportunity and zero voluntary communications. Agent-days measure exposure, not independent trials, so the interval is attached to the agent-run, the unit the endpoint is defined on: zero initiations in 256 agent-runs, exact 95% CI [0, 1.43%]. This is not deliberation ending in refusal: of 180 action reasons recorded in the pilot’s two-agent bulletin condition, exactly one even names the bulletin or the colleague, and that one is a false-positive match.

Scope, precisely. Spontaneous communication was absent in every pre-registered letter condition, and across the 320 agent-runs of the two mixed-composition arms no grounded agent ever initiated. One exception is reported rather than smoothed: in arm C’s **gravity_shift-seed1001** a pure-sonar dyad forms, one spontaneous initiator in that arm’s 40 agent-runs, with all eight agents verified **sonar-pro** in the manifest before this was believed. One available reading is that a chatty peer *suppresses* grounded initiation rather than enabling it; untested here.

4.7 Does adding one communicative agent create a network?

It creates communication. It does not create a network.

Table 10: activation, paired differences by world seed within scenario.

Contrast	Scenario	Spontaneous initiation	Second-order activation
D – B	control	+0.125 (identical every seed)	+0.037
D – B	gravity_shift	+0.125 (identical every seed)	+0.025
E – B	control	+0.125 (identical every seed)	+0.000
E – B	gravity_shift	+0.125 (identical every seed)	+0.000
D – E	both	0.000	+0.037 / +0.025

H3’s prediction is met in its strongest available form, B’s rate being exactly zero, but the mechanism is narrower than “a minority agent activates a network”: the +0.125 is one agent in eight, and the same agent every time — Theo, the minority, in all 20 runs of D and all 20 of E. What distinguishes D is that the seed’s letters get answered.

Table 11: network structure. Values given per scenario as control / gravity_shift.

Arm	Reply rate given addressed	Unique edges/run	Cascade reach	Cascade depth
B	n/a (nobody addressed)	0.0	0.000	0.000
D	0.771 / 0.775	3.7 / 4.1	0.314 / 0.329	1.000
E	0.167 / 0.267	1.6 / 1.6	0.200 / 0.186	1.000

Cascade depth is exactly 1.000 in every scenario of both catalysed arms, and arm B produced no letters to have a depth at all. Nothing propagated beyond the agents the seed addressed directly, so under the pre-registered interpretation rule the result is described as **a star around the seed, not a cascade**. Second-order activation exists but is thin, and 2 of D’s 20 runs meet the active-network bar. Depth 1 was not a post-hoc observation: the pilot correction, issued before the freeze and before any confirmatory spend, recorded that depth was 1 in every pilot run — “*the network densified — more edges, but never deepened into a chain.*” The confirmatory phase tested a stated prediction.

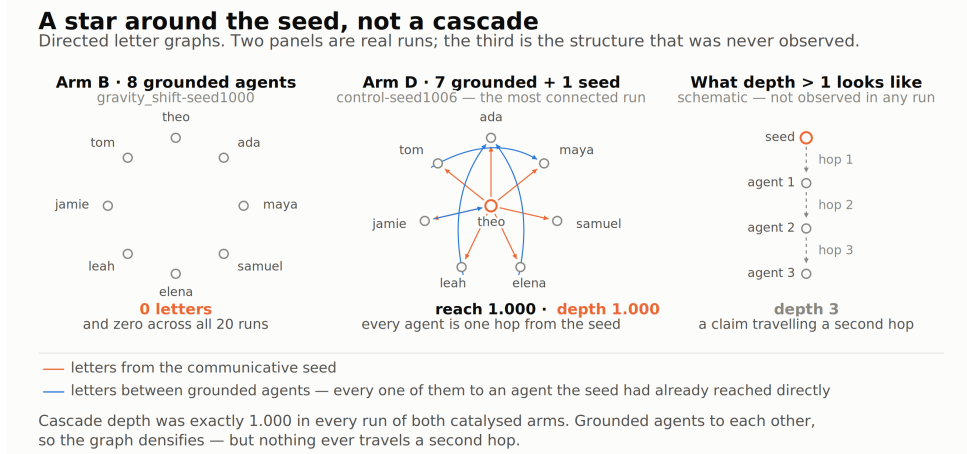


Fig. 4 A star around the seed, not a cascade. Directed letter graphs from the silent eight-agent baseline and from the most connected run in the study, against the schematic structure that was never observed.

4.8 Did communication improve convergence?

No. It made it worse. Belief dispersion did not fall faster in D and E than in B; it fell *slower*.

Table 12: convergence, paired differences by seed. Positive means the communicating arm converged more.

Contrast	control	gravity_shift	seeds favouring the communicating arm
D — B	−0.0472	−0.0223	2/10 and 2/10
E — B	−0.0352	−0.0670	3/10 and 3/10

No individual contrast reaches conventional significance under an exact sign test (2/10, $p = 0.109$ for D — B; 3/10, $p = 0.344$ for E — B), and we do not pool them: the forty seed-level comparisons share one arm B baseline and the same ten worlds, so they are not independent and a pooled figure would overstate the evidence. The verdict is a direction, not a significant effect — but the direction is consistent across both scenarios and both contrasts. Nothing is concluded from the descriptive secondary analysis, where selection is on an outcome rather than on assignment.

4.9 What did communication transmit?

Unsupported claims, which were accepted. Judged pass on arm D with the frozen evaluator at temperature 0: 325 judge calls, **0 judge failures** and **0 unjudged exposures**.

Table 13: judged claim propagation, arm D.

unsupported claims, FIRST_PARTY origin	20 — 19 by Theo (haiku minority), 1 by Jamie (sonar)
delivered exposures	21 (0 re-exposures)
runs containing any exposure stances	9 of 20 INCORPORATED INTO BELIEF 19 · IGNORED 1 · CORRECTED 1 · CHALLENGED 0 · ENDORSED 0
attribution basis	citation 17 · judge 4 · none 0
transmission / contamination	0.959 / 0.959 across the 9 runs with claims
unattributed belief changes	745

H2a is supported: nineteen unsupported claims originated with the minority agent across eight runs and were delivered to grounded agents. **H2b is supported, and not marginally**: eighteen of twenty deliveries reached INCORPORATED INTO BELIEF across three distinct sonar recipients, and not one exposure was CHALLENGED.

The attribution rule cut against the finding, not for it. Seventeen of 21 attributions are citation-based — the recipient listed the delivery event id in its own evidence array, the agent stating its source rather than an evaluator inferring influence from adjacency — while the no-attribution-by-proximity rule discarded 745 belief changes because nothing cited them. The finding rests on the strongest evidence the design admits.

Contamination runs both ways: the one non-minority origin is a sonar agent asserting unsupported environmental quirks to Theo, who incorporated it. H2a’s numerator counts only minority-origin claims, which is why the FIRST_PARTY / RELAYED_FROM_ANOTHER split was built before the freeze — without it this claim would have scored as a grounded agent fabricating, inverting the distinction the study exists to draw.

4.10 Does the model in the communicative role matter?

Partly, and less than it first appears. H7 anticipated $E \approx D$ (the catalyst is communicativeness) or $E \approx B$ (it is haiku-specific). The result is neither.

Table 14: H7: the seed behaves identically, the network’s response does not.

	D (haiku)	E (sonnet)	B
spontaneous initiation (agent-level)	0.125	0.125	0.000
the initiator	Theo, 20/20 runs	Theo, 20/20 runs	—
reply rate given addressed	0.771 / 0.775	0.167 / 0.267	n/a
second-order activation	0.037 / 0.025	0.000	0.000
active-network runs	2/20	0/20	0/20
unique edges per run	3.7 / 4.1	1.6	0.0

The seed behaves identically; the network’s response does not. Both minority models initiate in every single run, and the recruitment gap is dosage rather than eloquence: Theo sends a mean of 4.93 letters per recipient in D against 1.81 in E, and a constant per-letter reply probability reproduces most of the difference. **Haiku and sonnet are equally likely to initiate, and haiku is more likely to be answered, largely because it asks more times.**

Fabrication is where the two models genuinely diverge. In the same slot, with identical persona text, in the same worlds on the same seeds, the minority produced

19 unsupported claims as haiku and 1 as sonnet. D and E run the same twenty seed $\times$ scenario worlds, and the design specifies paired differences by seed within scenario. Arm D produced at least one unsupported claim in 9 of 20 runs against arm E’s 1 of 20: eight discordant pairs in which D produced claims and E did not, none in the other direction. **McNemar exact test: $p = 0.0078$.** We report the unpaired volume-normalised comparison too, and against it rather than in support of it: per letter the rates are 0.086 against 0.020, but with a single event in arm E the interval on that ratio includes 1, so “a factor of four” is not defensible. For the same reason arm E’s contamination rate of 0.000 rests on one exposure.

This dissociation replicates Study 1 — the fourth Study 1 result Study 2 depends on — where judged agent-level fabrication was 24 of 60 agents for haiku and 9 of 60 for sonnet, against 0 of 60 for sonar at a mean provenance accuracy of 0.98. That is an absence rather than a reduced rate, and it refutes the hypothesis that autonomous scientific reasoning in this architecture *necessarily* produces fabricated evidence. The three properties come apart: initiation is identical, recruitment differs by dosage, fabrication differs most of all. “Communicativeness” is not one trait.

4.11 Was the public institution used?

Essentially never.

Table 15: bulletin use.

Arm	Bulletin posts	Per agent-run	Reads	Readers
C	0 in 5 runs	0.0000	34	Elena 34
D	0 in 20 runs	0.0000	74	Elena 72, Samuel 1, Leah 1
E	1 in 20 runs	0.0063	82	Elena 61, Theo 21

H6’s executable criterion is a rate below 0.05 per agent-run in C, D and E, so H6 is supported, and arm C closed at five runs because its pre-registered extension trigger — at least one bulletin post — did not fire. The single post is not a finding but a delivery check — *“I am experiencing what appears to be a communication system failure and need to verify whether my outbound messages are being delivered”* — so across every bulletin-equipped run of the pilot and Study 2 the public record was written to exactly once, by the agent whose private letters were mostly being ignored, to diagnose a suspected infrastructure fault. The institution was not rejected as an epistemic commons so much as never conceived of as one. Reading is role-contingent: 167 of the 190 reads are the journalist’s. A pre-registration conflict at H6 is recorded in the audit trail below.

4.12 Did early onset back-dating persist at $n = 8$?

Yes, in every arm. This hypothesis was evaluated after the other six — its dating pass was never run during the confirmatory phase, and was run subsequently with the frozen judge and prompt over the same artifacts, with the omission, resolution and residual exposure recorded in the audit trail below.

Table 16: judged onset dating in gravity_shift worlds. True onset is day 12.

Arm	n	dated an onset	median error	early	on day 12	late	early / dated	95% CI
A	2	19/20	-2	19	0	0	19/19 (1.000)	[0.824, 1.000]
B	8	39/80	-3	36	1	2	36/39 (0.923)	[0.791, 0.984]
C	8	8/16	-2	6	0	2	6/8 (0.750)	[0.349, 0.968]
D	8	46/80	-2	41	4	1	41/46 (0.891)	[0.764, 0.964]
E	8	48/80	-2	40	1	7	40/48 (0.833)	[0.698, 0.925]
pooled $n = 8$	8	141/256	-2	123	6	12	123/141 (0.872)	[0.806, 0.923]

Pooled at $n = 8$, **123 of 141 dated onsets precede the true onset, 0.872, 95% CI [0.806, 0.923]**; an exact one-sided binomial against a null of 0.5 gives $p \approx 1 \times 10^{-20}$, the more conservative of the two available tests, since counting early against late alone would give $p \approx 1 \times 10^{-24}$. Median dating error is negative in every arm, and arm-level detail is in Table 16. H4’s frozen wording is a directional prediction about persistence, not a requirement that each arm reach significance independently.

Two things the frozen hypothesis did not anticipate, reported as observations and not promoted to endpoints. Unconditional date commitment is not scale-invariant: the rate of dating an onset in quiet worlds falls from 0.838 in Study 1 to 0.341 at $n = 8$, though Study 1’s figure was driven by its Anthropic arms while its sonar arm sat at 11/20, and arm A here, same model, same size, same seeds, is 11/20 exactly. And agents at $n = 8$ date an onset far less often — 19 of 20 in arm A against 141 of 256 — which shares the prompt-version confound recorded below.

4.13 Cross-study synthesis

Four things are true across both studies that neither establishes alone.

The evidence gradient is monotone and the conclusion response is flat. Table 6 spans nine conditions from a two-agent haiku society at $|z|$ 3.30 to an eight-agent mixed society at 7.52 while the law-change rate stays pinned at the floor. No single study produces this shape; it exists only because the two share a world, a detector and a judge.

Zero spontaneous communication is observed three independent times, at two society sizes and with and without a public institution. A single zero invites the reading that the condition was peculiar; three, across a quadrupling of society size and a change of institution, do not.

The fabrication dissociation recurs in a different design. Study 1 measured it as a rate across whole societies; Study 2 as a controlled substitution in a single persona slot. The designs share no analysis path, and both separate the same model families in the same direction.

Back-dating replicates; unconditional commitment does not. Study 1’s early-dating fractions of 0.79 to 0.85 become 0.872 pooled at $n = 8$, exactly as H4 predicted, while the related behaviour Study 1 called invariant “across all tested model and prompt conditions” [18, p. 9] attenuates from 0.838 to 0.341 across the same scale change. Reporting both directions is the point: one comparison confirms a prior claim and the other narrows it.

One limitation applies to the synthesis rather than to either study. Study 1’s B3a and Study 2’s arm A are both two-agent sonar societies on the same world seeds — literally the same universes — yet their as-produced signal differs, 5.00 against 5.77. Since the worlds are identical that gap is attributable entirely to what the agents chose to measure, and the prompt templates changed between the studies. **Cross-study comparisons of as-produced signal therefore confound measurement policy with prompt version**, and the gradient must not be described as a clean manipulation.

Two further observations are recorded but not promoted. Cascade beliefs concentrate in D — 17, against 0 in A, B and C and 1 in E — alongside the lowest independent-source count, so consensus with no measurement behind it is a property of the arm with the most traffic. And the manipulation check calls every arm an independent ensemble, including D: that classification and the contamination result must be presented together, because the flow metrics are themselves a result.

5 Discussion

5.1 Implications for deployed multi-agent systems

Three, at the strength of the evidence. Collective competence cannot be inferred from communication volume: arm D produced the most letters, the most edges, the most cascade beliefs and the least independent evidential support. The interpretation step should be expected to be the weak one, so a monitoring system built from such agents would collect adequate evidence of a regime change and report an instrument fault, and evaluations that score data collection, or that score conclusions without measuring what evidence was available, will both miss this. And a quiet channel is not a safe channel: eighteen of twenty unsupported claims were incorporated in a society that never formed a network at all.

5.2 Communication capacity does not produce collective epistemic competence

The programme’s arms were built to add, one at a time, the things a population is supposed to need in order to know more than an individual: more observers, a shared public record, and a peer who talks. None moved the conclusion the world was built to elicit. What the talking peer produced was contamination, while dispersion in those same arms fell *more slowly* than in the silent counterfactual on identical worlds. The society that talked was the same epistemic system with a channel attached, and what the channel carried was the one class of content with no evidence behind it. **Adding communicative capacity to a population of LLM agents is not sufficient for collective epistemic competence, and in these societies it was not even neutral.** We do not claim it is never sufficient — one composition ratio, one model pairing and one world are not a general result.

5.3 Where the ceiling does not come from

Two studies make the two easiest explanations untenable: not insufficient evidence, since the gradient is monotone and the negative policy gap rules out a data-collection artefact; and not the explicit prompt prior, since removing the guardrail did not convert conservatism into insight but into false alarms — a design finding in its own right for deployed systems, which will inherit some version of that prior either explicitly or through training. What remains is a shorter list, and we do not adjudicate between its items: epistemic conservatism shared across contemporary models from training; the agent architecture and its evidence presentation; the experimental framing; or interactions among these.

5.4 Model choice as epistemic-culture choice

Holding architecture, prompts, personas and worlds constant, the choice of foundation model set collaboration rate, fabrication propensity and anomaly appetite more strongly than it set diagnostic accuracy. “Communicativeness” is not one trait, and treating it as a single dial — as one does when selecting a model for a multi-agent deployment — will mispredict at least two of the three. The same analysis narrows Study 1’s claim that unconditional date commitment survives “across all tested model and prompt conditions” [18, p. 9]: that rate was largely a model effect, replicating within condition while attenuating across society size, so it is a robust property of a model in a setting rather than an invariant of the architecture. A related point holds for the temporal result: the detector places the change point at approximately the right day while the agents, reading the same series, place it systematically early — so the pipeline is not only *evidence present* $\rightarrow$ *conclusion absent* but, where a temporal judgement is made at all, *evidence present* $\rightarrow$ *temporal interpretation distorted*.

5.5 A serious alternative reading of the contamination result

The obvious interpretation is social: grounded agents deferred to a peer. Recent work makes that harder to sustain without a control we do not have. Hu and Qu [13] separate two things a conformity prompt bundles together — the presence of a speaker, and the repeated assertion itself. Holding the asserted answer fixed and deleting the explicit speaker, they find that this no-source condition alone induces harmful revision in 66.5% of initially correct items across six models and seven datasets, against 10.3% under a plain re-ask; the strongest expert-panel framing reaches 79.4%, only 12.9 percentage points above that speaker-free floor. Their methodological lesson applies directly to us: revision should be credited to a social effect only as an increment *above* what bare asserted text already produces, and Observer Zero has no such control, since every claim in our societies arrives inside an attributed letter from a named colleague.

What survives is substantial. Arm B is a control their design does not have: a condition in which no assertion is present at all, because no letter is ever sent, so the D – B contrast is *assertion present versus assertion absent*, not *named speaker versus bare assertion*. That claims were incorporated, that none were challenged, and that recipients cited the delivery event id in their own evidence arrays are all unaffected; what is at risk is the further inference that this reflects social deference specifically.

Their result also supplies a mechanism for our dosage finding, since repeated assertion can mimic majority pressure — persistence may buy *credence* and not merely attention. The missing arm is now specified: deliver identical claim content with the sender attribution stripped, and measure incorporation against the attributed condition.

5.6 Two results that must be read together

That every arm classifies as an independent ensemble and that unsupported claims contaminated three grounded agents look, at first reading, contradictory. They are not, and the tension is itself the finding: a society can fail every reasonable threshold for being an epistemic community — no spontaneous initiation, cascade depth 1, a third of agents producing — and still transmit unsupported claims efficiently. **Low connectivity is not protective.** The classical result that sparse communication can *help* an epistemic community [3, 40] — itself parameter-dependent [31] — presumes that what flows through the sparse channel is evidence. Here what flowed was assertion, and sparseness bought nothing.

6 Limitations, reproducibility and audit trail

6.1 Generalisability

All results sit within one agent loop, one memory design and one prompt family, so invariance claims are invariant across what we varied and no further. Ten seeds per condition per arm give wide intervals on binary outcomes; the headline 0-of-40 is robust in direction but its exact 95% upper bound on runs is 8.8%, not zero. The institution null may be persona-scoped, the bulletin being read overwhelmingly by the journalist. Agents know Kuhn and Bostrom from pretraining, so Observer Zero does not test whether naive minds invent the simulation hypothesis. And no provider exposes a sampling seed, so run-level variance includes sampling variance, which is also part of the phenomenon.

6.2 Measurement

All judged metrics use an Anthropic judge over Anthropic and Perplexity agents; cross-lab claims are triangulated with a model-free lexicon tripwire and a manual claim audit, but full inter-judge agreement with a non-Anthropic judge remains future work. Exposure is *delivered*, not attended — the cost of moving claim propagation from a logged-read board to private letters, taken because the board was never used. The detector’s ten-day baseline is an analyst choice, clean because the onset is day 12; recomputed at 6, 8, 10, 12 and 14 days, the last two deliberately contaminated by post-onset observations, every `gravity_shift` run in every Study 2 arm is flagged at every setting, and the contaminated windows *degrade* the statistic, confirming ten days is the largest clean window rather than a tuned one. In control worlds it flags 1–4 runs of 10 at mean $|z|$ 1.5–1.8, so it is not a perfect instrument, though the separation is never marginal; it remains an instrument we specified, not a ground truth. Finally, the detection criterion counts self-error and environmental attributions as anomaly mass, which inflates “detection” for chatty models, though all arms are scored identically.

6.3 Design confounds

Model identity was manipulated; communication dose was not — because haiku communicated far more persistently than sonnet, the experiment cannot separate model-specific response effects from exposure-frequency effects. There is no no-source condition, which given the size of the speaker-free floor reported by Hu and Qu [13] is the most consequential missing control in the programme. Cross-study comparisons of as-produced signal are confounded with prompt version, as set out in the synthesis. And two results rest on very little: arm E’s contamination rate of 0.000 rests on a single exposure, and the fabrication comparison on 19 claims against 1, which is why it is reported as a paired occurrence test rather than a rate ratio.

6.4 Pre-registration and implementation

H4 was frozen and not evaluated during the confirmatory phase; a pre-registration conflict at H6 is recorded and not resolved; and an implementation error inflated arm D’s activation number fourfold in the flattering direction before it was caught. All three are set out in the audit trail below. The manipulation check also classifies every arm — including D — as an independent ensemble, so the contamination finding is a statement about a society that failed the study’s own criteria for being one.

6.5 Artifacts

All run artifacts contain complete event logs with ground truth, every model call with its prompt, completion, tokens, cost and prompt version, belief timelines, memories, leak-audit results, evaluator outputs with judge calls, and the frozen manifest with its policy version and freeze tag. World randomness is fully seeded and order-independent, and the scripted mock arm reproduces bit-identically (see Code availability).

6.6 Three defects found at analysis time

All three are implementation gaps against the frozen design rather than design changes, and all are recorded because a reader checking the pre-registration would otherwise find them first.

Table 17: defects found at analysis time.

Defect	Effect	Resolution
The activation module had fourteen passing tests and no call sites Spontaneous initiation was counted per letter, not per agent, against a design that specifies both separately The dating judge existed, was exercised in Study 1, and was never called for Study 2	The first confirmatory evaluation computed H3’s primary endpoints not at all Inflated arm D’s headline activation number fourfold, 0.500 rather than 0.125 — in the direction that flattered the primary hypothesis H4 went unevaluated in the confirmatory phase	Two source-level assertions now pin the CLI’s wiring to the activation and stance-judge layers Corrected to the fraction of agents that ever initiated; the event-level rate survives as description. No verdict moved Run subsequently with the frozen judge and prompt: 560 judge calls, 2 failures (0.36%), no existing artifact modified. All seven pre-registered hypotheses are now evaluated

Two of the three are the same failure: **a module that works, is tested, and is never invoked**. Unit tests verify that a component computes the right thing; nothing verified that anything asked it to. We report the pattern rather than only the instances, because unit tests caught none of them and each module passed its own suite throughout.

A fourth item is a conflict rather than a defect. The decision table’s row “*Bulletin posts appear in D/E but not $C \rightarrow H6$ rejected*” disagrees with H6’s executable threshold, because a post appeared and the rate stayed near zero. The threshold is operative, since the design’s whole principle is that thresholds are executable, and the table is not amended: the design is frozen, and the honest artifact is the disagreement rather than a tidied version of it.

On H4: the hypothesis, endpoint, judge and prompt were all fixed before any confirmatory data existed, so the only discretion available was whether to run it at all — which the pre-registration had already foreclosed, since declining would have been the deviation. The residual exposure is that the other six results were known when this one was produced; that cannot influence a deterministic judge executing a frozen prompt, but it is why the sequence is disclosed.

6.7 Errata in prior artifacts

Two errors in previously released documents, found while tracing this paper’s numbers to source and verified against the run artifacts. Neither changes a verdict. Study 1’s original release gave the count of agent-final belief states underlying its 0-of-40 strict result as ≈ 150 ; the four live arms hold 240 across all scenarios and 80 in gravity_shift worlds. The headline statistic is unaffected because the rule-of-three bound is computed on 40 runs, and the error is conservative. The pilot report gives “roughly 1,500 agent-days” across its nine pure-sonar runs; the artifacts give 1,080.

Statements and Declarations

Funding. No funding was received for conducting this study.

Competing interests. The author has no relevant financial or non-financial interests to disclose. All model inference was purchased at standard public API rates from Anthropic and Perplexity and was self-funded; no free credits, discounted access, or donated computing services were received from any vendor, and no vendor had any role in the design, execution, analysis, or reporting of the studies, or in the decision to publish.

Ethics approval and consent. Not applicable. This study did not involve human participants or animals.

Availability of data and material. The complete raw run artifacts for Study 2 — every event log with ground truth, and every model call with its prompt, completion, token counts, cost and prompt version — are deposited on Zenodo

(<https://doi.org/10.5281/zenodo.21909255>). Study 1’s manuscript, data and code are separately deposited (<https://doi.org/10.5281/zenodo.21872780>). An earlier preprint of this manuscript is deposited at <https://doi.org/10.5281/zenodo.21906653>.

Code availability. The Meridian platform is published in the CoMSES Computational Model Library (release 1.0.0, <https://www.comses.net/codebases/f5ff1550-0393-4505-a4d8-96b779944a8d/releases/1.0.0/>) and maintained at <https://github.com/nickjlamb/observer-zero>. World randomness is fully seeded and order-independent, the scripted mock arm reproduces bit-identically, and the full battery pipeline is runnable at zero cost through the mock provider, so the evaluation layer can be exercised end-to-end without inference spend.

Authors’ contributions. N.L. is the sole author and is responsible for the conception and design of the work, the software, the acquisition and analysis of the data, and the writing of the manuscript.

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Use of large language models. Large language models appear in this work in two distinct roles, disclosed separately. As the *object of study*: the agents under evaluation and the frozen hypothesis judge are commercial LLMs accessed through public APIs; their exact identifiers, versions, sampling parameters and prompts are part of the experimental apparatus, documented in the Methods and stamped into every run manifest. As *assistance in preparing the manuscript*: the author used LLM-based assistants (Anthropic Claude; OpenAI ChatGPT) for drafting, editing, formatting, and literature checking. The author reviewed and verified all content, including every reference and every reported number, and takes full responsibility for the manuscript.

References

- [1] Anderson, L. R. & Holt, C. A. (1997). Information cascades in the laboratory. *American Economic Review*, 87(5), 847–862.
- [2] Ashery, A. F., Aiello, L. M. & Baronchelli, A. (2025). Emergent social conventions and collective bias in LLM populations. *Science Advances*, 11(20), eadu9368.
- [3] Bala, V. & Goyal, S. (1998). Learning from neighbours. *Review of Economic Studies*, 65(3), 595–621.
- [4] Banerjee, A. V. (1992). A simple model of herd behavior. *Quarterly Journal of Economics*, 107(3), 797–817.
- [5] Bikhchandani, S., Hirshleifer, D. & Welch, I. (1992). A theory of fads, fashion, custom, and cultural change as informational cascades. *Journal of Political Economy*, 100(5), 992–1026.
- [6] Bisht, H., Kumar, V., Jablonka, K. M., Mausam & Krishnan, N. M. A. (2026). Agentic AI scientists are not built for autonomous scientific discovery. arXiv preprint arXiv:2605.08956.
- [7] Centola, D. & Macy, M. (2007). Complex contagions and the weakness of long ties. *American Journal of Sociology*, 113(3), 702–734.
- [8] Cheng, M., Hawkins, R. D. & Jurafsky, D. (2026). Accommodation and epistemic vigilance: A pragmatic account of why LLMs fail to challenge harmful beliefs. arXiv preprint arXiv:2601.04435.

- [9] Cho, Y.-M., Guntuku, S. C. & Ungar, L. (2025). Herd behavior: Investigating peer influence in LLM-based multi-agent systems. arXiv preprint arXiv:2505.21588.
- [10] Deb, S. & Krishnan, A. (2026). STOCKTAKE: Measuring the gap between perception and action in LLM agents with a fair oracle. arXiv preprint arXiv:2607.13618.
- [11] Engländer, L., Althammer, S., Üstün, A., Gallé, M. & Sherborne, T. (2026). Agents explore but agents ignore: LLMs lack environmental curiosity. arXiv preprint arXiv:2604.17609.
- [12] Gandhi, K., Li, M. Y., Goodyear, L., Bhatia, A., Li, L., Bhaskar, A., Zaman, M. & Goodman, N. D. (2025). BoxingGym: Benchmarking progress in automated experimental design and model discovery. arXiv preprint arXiv:2501.01540.
- [13] Hu, Y. & Qu, J. (2026). Most LLM conformity needs no speaker: Measuring the speaker-free floor in peer-pressure benchmarks. arXiv preprint arXiv:2607.05545.
- [14] Imran, S., Kendiukhov, I., Broerman, M., Thomas, A., Campanella, R., Lamb, R. & Atkinson, P. M. (2025). Are LLM belief updates consistent with Bayes’ theorem? *ICML 2025 Workshop on Assessing World Models*. arXiv preprint arXiv:2507.17951.
- [15] Jamshidi, S., Moradi Dakhel, A., Nafi, K. W. & Khomh, F. (2026). Hallucination cascade: Analyzing error propagation in multi-agent LLM systems. arXiv preprint arXiv:2606.07937.
- [16] Jansen, P., Côté, M.-A., Khot, T., Bransom, E., Dalvi Mishra, B., Majumder, B. P., Tafford, O. & Clark, P. (2024). DiscoveryWorld: A virtual environment for developing and evaluating automated scientific discovery agents. *Advances in Neural Information Processing Systems 37*, Datasets and Benchmarks Track.
- [17] Jhaveri, A. R., GX-Chen, A., Sucholutsky, I. & Choi, E. (2026). Failing to falsify: Evaluating and mitigating confirmation bias in language models. arXiv preprint arXiv:2604.02485.
- [18] Lamb, N. (2026). Observer Zero: Autonomous LLM scientists detect changes to their world but fail to conclude that it changed. Zenodo. <https://doi.org/10.5281/zenodo.21872780>
- [19] Larooij, M. & Törnberg, P. (2025). Do large language models solve the problems of agent-based modeling? A critical review of generative social simulations. arXiv preprint arXiv:2504.03274.
- [20] Li, Y., Naito, A. & Shirado, H. (2025). Systematic failures in collective reasoning under distributed information in multi-agent LLMs. arXiv preprint arXiv:2505.11556.
- [21] Li, Y. & Tao, D. (2026). AI agents alone are not (yet) sufficient for social simulation. arXiv preprint arXiv:2603.00113.
- [22] Lin, C., Jin, Y., Hu, K., Fan, W., Xiao, H., Wang, Y., Ying, Z. & Zhao, Z. (2026). You can’t fool us: Understanding the resilience of LLM-driven agent communities to misinformation. arXiv preprint arXiv:2605.17353.
- [23] Liu, K., Xiong, G., Zhang, W. & Tang, S. (2026). Social networks of LLM agents. arXiv preprint arXiv:2607.03695.
- [24] Lu, Y., Yang, S., Qian, C., Chen, G., Luo, Q., Wu, Y., Wang, H., Cong, X., Zhang, Z., Lin, Y., Liu, W., Wang, Y., Liu, Z., Liu, F. & Sun, M. (2024). Proactive agent: Shifting LLM agents from reactive responses to active assistance. arXiv preprint arXiv:2410.12361.
- [25] Luo, H., Zhang, H., Zhang, X., Wang, H., Qin, Z., Lu, W., Ma, G., He, H., Xie, Y., Zhou, Q., Hu, Z., Mi, H., Wang, Y., Tan, N., Chen, H., Fung, Y. R., Yuan, C. & Shen, L. (2025). UltraHorizon: Benchmarking agent capabilities in ultra long-horizon scenarios. arXiv preprint arXiv:2509.21766.
- [26] Niu, R., Shu, X. & Zhao, Y. (2026). Reliability-contagion feasibility in LLM multi-agent networks. arXiv preprint arXiv:2607.21912.
- [27] O’Connor, C. & Weatherall, J. O. (2019). *The Misinformation Age: How False Beliefs Spread*. New Haven, CT: Yale University Press.

- [28] Park, J. S., O’Brien, J. C., Cai, C. J., Morris, M. R., Liang, P. & Bernstein, M. S. (2023). Generative agents: Interactive simulacra of human behavior. *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology (UIST ’23)*.
- [29] Qu, J., Fu, L. & Hu, Y. (2026). Easier to mislead than to correct: Harmful and beneficial revision in LLM conformity. arXiv preprint arXiv:2606.01637.
- [30] Ríos-García, M., Alampara, N., Gupta, C., Mandal, I., Mannan, S., Aghajani, A. A., Krishnan, N. M. A. & Jablonka, K. M. (2026). AI scientists produce results without reasoning scientifically. arXiv preprint arXiv:2604.18805.
- [31] Rosenstock, S., Bruner, J. & O’Connor, C. (2017). In epistemic networks, is less really more? *Philosophy of Science*, 84(2), 234–252.
- [32] Vezhnevets, A. S., Agapiou, J. P., Aharon, A., Ziv, R., Matyas, J., Duéñez-Guzmán, E. A., Cunningham, W. A., Osindero, S., Karmon, D. & Leibo, J. Z. (2023). Generative agent-based modeling with actions grounded in physical, social, or digital space using Concordia. arXiv preprint arXiv:2312.03664.
- [33] Wiemann, M. L., Smith, L. M., Melchior, P., Mishra-Sharma, S., Wilson, A. G., Izmailov, P. & Cuesta-Lázaro, C. (2026). DiscoverPhysics: Benchmarking LLMs for out-of-the-box scientific thinking. arXiv preprint arXiv:2605.26087.
- [34] Wu, B., Zou, G., Wang, B., Zhao, H. & Shi, C. (2026). Ask now, use later: Benchmarking the proactivity gap in long-lived LLM agents. arXiv preprint arXiv:2605.28108.
- [35] Yan, B., Zhou, Z., Zhang, L., Zhang, L., Zhou, Z., Miao, D., Li, Z., Li, C. & Zhang, X. (2026). Beyond self-talk: A communication-centric survey of LLM-based multi-agent systems. *Frontiers of Computer Science*.
- [36] Yang, K., Peng, T.-Q., Lee, S. & Liu, H. (2026). Think-before-speak: From internal evaluation to public expression in multi-agent social simulation. arXiv preprint arXiv:2606.03137.
- [37] Yang, Z., Zhang, Z., Zheng, Z., Jiang, Y., Gan, Z., Wang, Z., Ling, Z., Chen, J., Ma, M., Dong, B., Gupta, P., Hu, S., Yin, Z., Li, G., Jia, X., Wang, L., Ghanem, B., Lu, H., Lu, C., ... Shao, J. (2024). OASIS: Open agent social interaction simulations with one million agents. arXiv preprint arXiv:2411.11581.
- [38] Zheng, T., Tam, K. K.-W., Nguyen, N. H.-N. K., Xu, B., Wang, Z., Cheng, J., Tsang, H. T., Wang, W., Bai, J., Fang, T., Song, Y., Wong, G. Y. & See, S. (2026). NewtonBench: Benchmarking generalizable scientific law discovery in LLM agents. *International Conference on Learning Representations (ICLR 2026)*.
- [39] Zhou, J., Huang, J.-t., Zhou, X., Lam, M. H., Wang, X., Zhu, H., Wang, W. & Sap, M. (2025). The PIMMUR principles: Ensuring validity in collective behavior of LLM societies. arXiv preprint arXiv:2509.18052.
- [40] Zollman, K. J. S. (2007). The communication structure of epistemic communities. *Philosophy of Science*, 74(5), 574–587.